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Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur
Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process
an Airbag Operation

Phase 7: Hardware Safety Requirements Version: 1.00

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1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	11:00	Zoom	ALL
Bericht:	<u>none</u>		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklärung / Explanation
ACU	Airbag Control Unit the ECU responsible for crash detection and airbag deployment control
ADC	Analog-to-Digital Converter converts analog sensor voltage to a digital integer value
ARM Cortex-M3	32-bit RISC processor core used in the Arduino Due (SAM3X8E)
ASIL	Automotive Safety Integrity Level A to D, D being highest requirement (ISO 26262)
CAN	Controller Area Network automotive serial communication bus standard (ISO 11898)
ECU	Electronic Control Unit embedded computing module in a vehicle
ESD	Electrostatic Discharge overvoltage event that can damage electronic components
EMI	Electromagnetic Interference unwanted electrical noise affecting circuit operation
GPIO	General Purpose Input/Output configurable digital pin on a microcontroller
HW	Hardware
ISO 26262	International standard for functional safety of road vehicle electrical/electronic systems
LDO	Low Drop-Out Regulator linear voltage regulator with low input-output voltage differential
MCU	Microcontroller Unit
MEMS	Micro-Electro-Mechanical System miniature sensor technology used for accelerometers and gyroscopes
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor used as switching element in output drivers
Safing Condition	A secondary, independent confirmation of crash event required before deployment is enabled (dual-point safety architecture)
SWR	Software Requirement functional requirement assigned to the software module
SW	Software
TVS	Transient Voltage Suppressor protection diode for input overvoltage clamping

4 Hardware Safety Requirements

This document defines the Hardware Safety Requirements for the Airbag ECU. The hardware must reliably process sensor signals, make deployment decisions, and communicate fault states within the tight timing constraints of a real-world crash event (typically less than 30 ms from impact to deployment).

4.1 Safety-Relevant Functions

The hardware must support the following safety-relevant system functions defined in the system concept.

Tabelle 2 Safety Relevant Functions

ID	Function	Reference
F1	Crash detection via acceleration, gyroscope, and pressure sensors	Airbag_ECU_Phase_2, 4.1 Identified Hazards, Table 2, H1, H2
F2	Airbag squib firing (simulated via digital output in prototype)	Airbag_ECU_Phase_3, 4.1 Safety Goals, Table 2, SG2
F3	Sensor monitoring and plausibility checking	Airbag_ECU_Phase_3, 4.2 Technical Safety Mechanisms, Table 3
F4	System self-test and diagnostics	Airbag_ECU_Phase_3, 4.2 Technical Safety Mechanisms, Table 3
F5	Fail-safe / fall-back on detected hardware faults	Airbag_ECU_Phase_3, 4.1 Safety Goals, Table 2, SG1
F6	CAN communication of safety-relevant states to vehicle ECU	Airbag_ECU_Phase_3, 4.1 Safety Goals, Table 2, SG1, SG2

4.2 Sensor Input Requirements and Thresholds

Three analog sensor channels provide the crash detection inputs.

Tabelle 3 Sensor Inputs and Thresholds

Sensor	Pin	ADC Range	Crash Threshold	Function
Acceleration Sensor (frontal crash)	A0	0-1023 (10-bit ADC)	≥ 700	Triggers frontal crash condition
Gyroscope Sensor (rollover / safing)	A1	0-1023 (10-bit ADC)	≥ 600	Required as safing channel (dual-point logic)
Pressure Sensor (side impact)	A2	0-1023 (10-bit ADC)	≥ 650	Triggers side crash condition
Baby/Passenger Seat Switch	D2	Digital Active LOW (INPUT_PULLUP)	LOW = baby seat	Suppresses passenger airbag deployment

4.3 Sensor Validity Monitoring Requirements

The hardware design must support plausibility checking of sensor outputs as required by safety function F3.

ADC value greater than 1020 or less than 2 = sensor short-circuit or open-circuit fault detected = safe state entry.

Multiple sensors simultaneously at maximum reading = implausible condition = flag for software plausibility check.

Gyroscope reading absent while acceleration indicates crash = safing channel fault = block deployment.

4.4 Output Driver Requirements

Tabelle 4 Output Driver Requirements

Output Function		Pin	Logic Active	Description
Driver Deployment	Airbag	D8	HIGH (3.3 V)	Activates when valid crash + safing condition TRUE
Passenger Deployment	Airbag	D9	HIGH (3.3 V)	Activates only when crash valid AND baby seat NOT detected
Warning Lamp (Fault Indicator)	(Fault Indicator)	D13	HIGH (3.3 V)	Illuminated during safe state or detected fault condition

4.5 Safe State Requirements Fault Tolerance

The hardware safe state ensures that airbag deployment cannot occur accidentally when a hardware fault is detected. Safe state: Driver airbag output = LOW, Passenger airbag output = LOW, Warning lamp = HIGH, CAN fault frame transmitted.

Tabelle 5 Safe State Trigger Conditions

ID	Trigger Condition	Hardware Implication
FS01	ADC value out of valid range (less than 2 or greater than 1020)	Sensor open/short circuit; hardware fault block deployment
FS02	Output readback mismatch (commanded HIGH, reads LOW)	Output driver failure; hardware fault activates warning lam
FS03	CAN bus error detected (bus-off state)	Communication hardware fault enters safe state
FS04	Baby seat signal intermediate voltage (neither 0 V nor 3.3 V)	Seat switch hardware fault passenger airbag suppressed, warn
FS05	Watchdog timer expiry (MCU stall)	Hardware watchdog resets MCU; outputs go to default LOW (safe)

4.6 4.6 Hardware Requirements Traceability

Tabelle 6 Hardware Requirements Traceability

SWR ID	Software Requirement	Supporting Hardware Design Element
SWR01	Read acceleration sensor input	Analog input A0, 10-bit ADC, potentiometer simulation (production: MEMS accelerometer)
SWR02	Read gyroscope angular sensor input	Analog input A1, 10-bit ADC (production: MEMS gyroscope with signal conditioning)
SWR03	Read pressure sensor input	Analog input A2, 10-bit ADC (production: MEMS pressure sensor + low-pass filter)
SWR04	Read passenger seat digital input	Digital input D2 with INPUT_PULLUP; active-LOW logic
SWR05	Activate driver airbag output	Digital output D8 (production: high/low-side MOSFET squib driver)
SWR06	Activate passenger airbag (no baby seat)	Digital output D9 with D2 seat switch interlock logic
SWR07	Activate warning lamp during fault	Digital output D13 (active HIGH)
SWR08	Send crash/fault status via CAN	Arduino Due CAN0 controller + MCP2551 transceiver, ID 0x120
SWR09	Read CAN messages for diagnostic/reset	CAN0 receive interrupt handler; due can library
SWR10	Read back output status after activation	GPIO readback of D8/D9 after asserting; hardware feedback path in production
SWR11	Enter safe state if fault detected	Hardware watchdog, output default-LOW on reset, warning lamp D13
SWR12	Support test routines for module/integration testing	Serial/USB debug interface; Wokwi simulation platform for hardware-in-loop testing

5 Figure and table listing

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6 List of dependent documents

No.	Dokument name	Version	Date	File name
1	Hazard Analysis and ASIL Determination for Airbag ECU	V02	02.06.2026	Hazard Analysis and ASIL Determination for Airbag ECU V02.pdf
2	ECU Airbag System Architecture Documentation	V02	27.05.2026	ECU Airbag system architecture documentation V2.pdf
3	Airbag ECU Hardware Concept Design Report	V02	22.05.2026	AirbagECU_HW_Concept_Design_Report_V02.pdf

7 Literature

No. Reference

- [1] International Organization for Standardization (ISO), ISO 26262: Functional Safety Standard for Modern Road Vehicles. https://fscdn.rohm.com/en/products/databook/white_paper/iso26262_wpe.pdf
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8 Version history

Document name	date	Change log
Airbag_ECU_Phase_7	16.06.2026	v1.00 Initial version. Hardware Safety Requirements compiled from HW Concept Design Report V02 Sections 2, 5, 6, 9, 10.